

The sinking of the Titanic is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the Titanic sank after colliding with an iceberg, killing 1502 out of 2224 passengers and crew. That's why the name DieTanic. This is a very unforgettable disaster that no one in the world can forget.

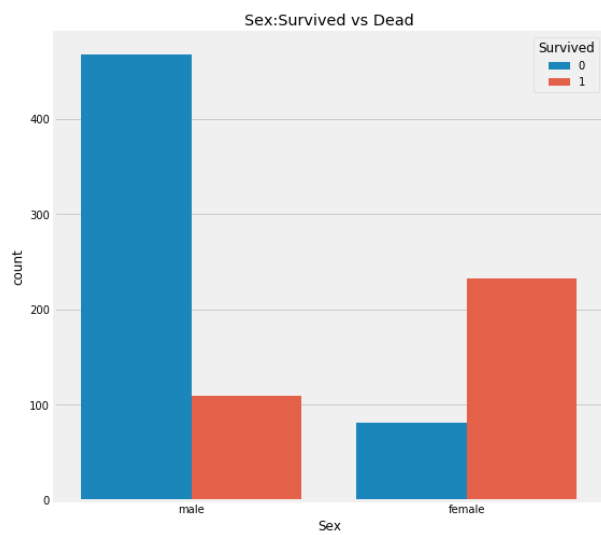
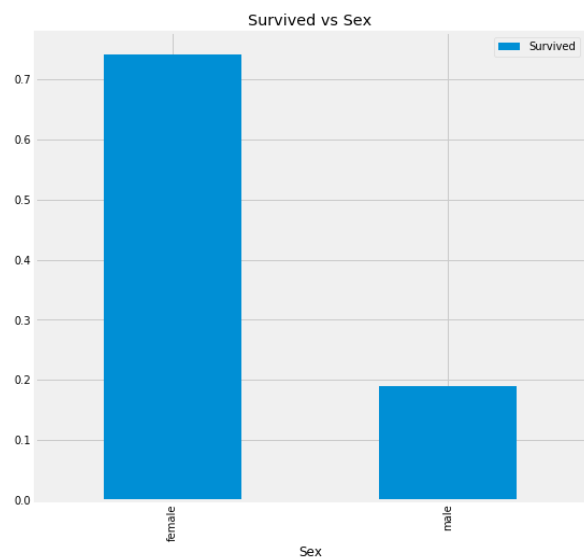
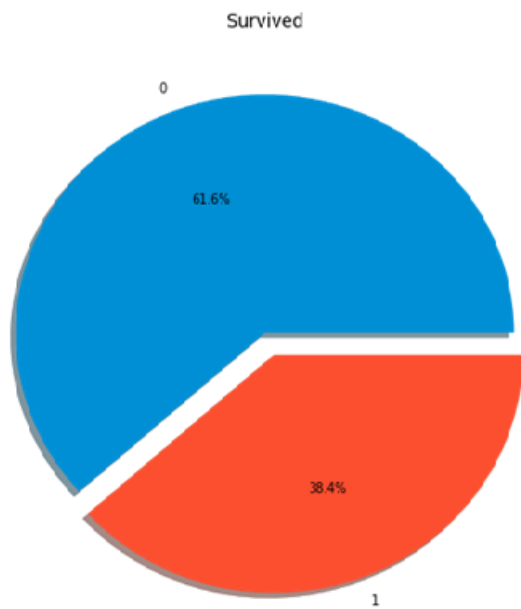
It took about \$7.5 million to build the Titanic and it sunk under the ocean due to collision. The Titanic Dataset is a very good dataset for beginners to start a journey in data science and participate in competitions in Kaggle.

The Objective of this document is to give an idea how is the workflow in any predictive modeling problem. How do we check features, how do we add new features and some Machine Learning Concepts. I have tried to keep the document as basic as possible so that even newbies can understand every phase of it.

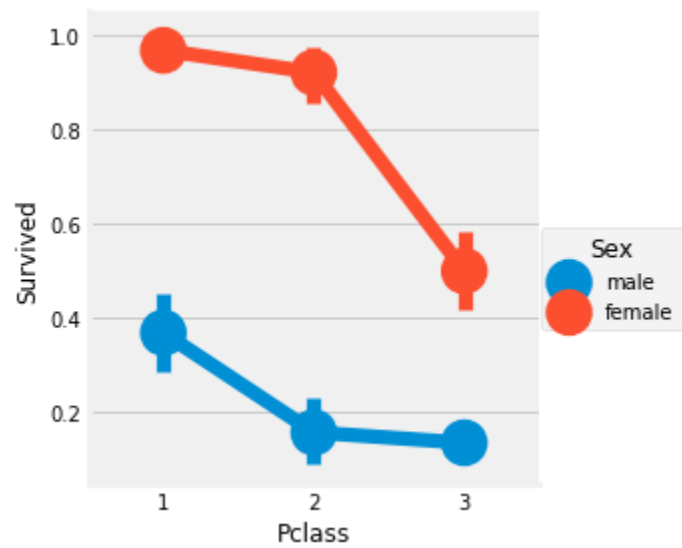
This document provides a comprehensive guide on performing Exploratory Data Analysis (EDA) and predictive modeling on the Titanic dataset. It covers data cleaning, feature engineering, visualization, and model building using various algorithms. The notebook aims to predict passenger survival rates using machine learning techniques and offers insights into the factors influencing survival. sibsp is the number of siblings/spouses aboard. Parch is the number of parents/children aboard. For embarked, C = Cherbourg, Q = Queenstown, S = Southampton.

Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr. James	male		0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075		S
9	1	3	Johnson, Mrs. Oscar W	female	27	0	2	347742	11.1333		S

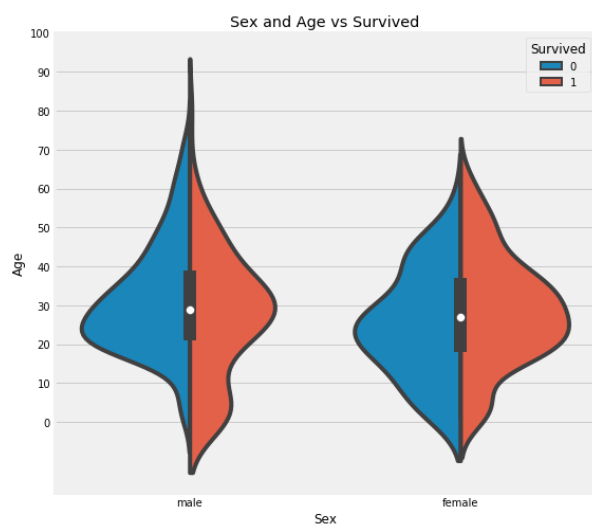
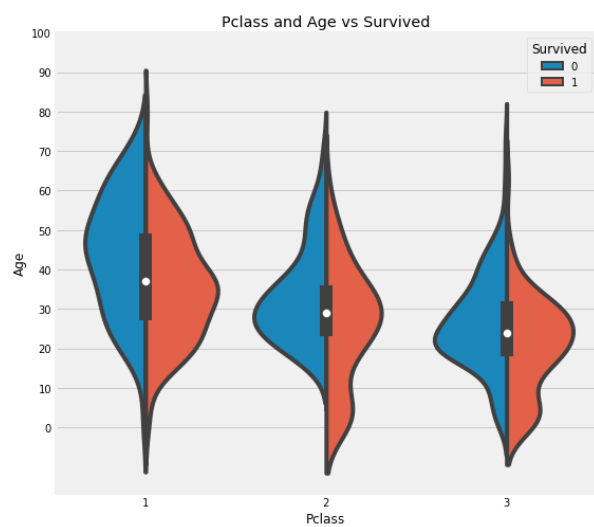
			(Elisabeth Vilhelmina Berg)								
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	C103	S
13	0	3	Saundercock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson, Mr. Anders Johan	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55	0	0	248706	16		S
17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125		Q
18	1	2	Williams, Mr. Charles Eugene	male		0	0	244373	13		S
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31	1	0	345763	18		S



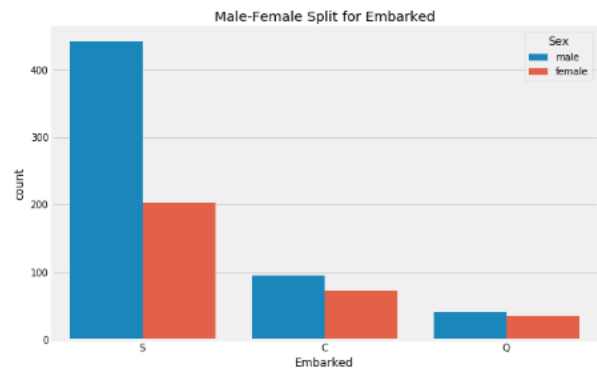
Survived	0	1	All
Pclass			
1	80	136	216
2	97	87	184
3	372	119	491
All	549	342	891

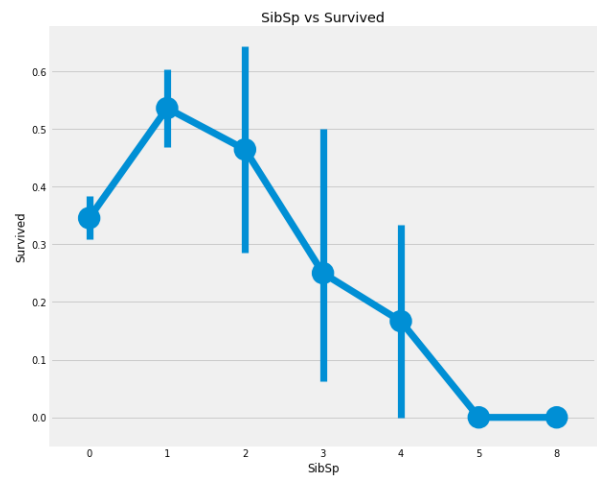


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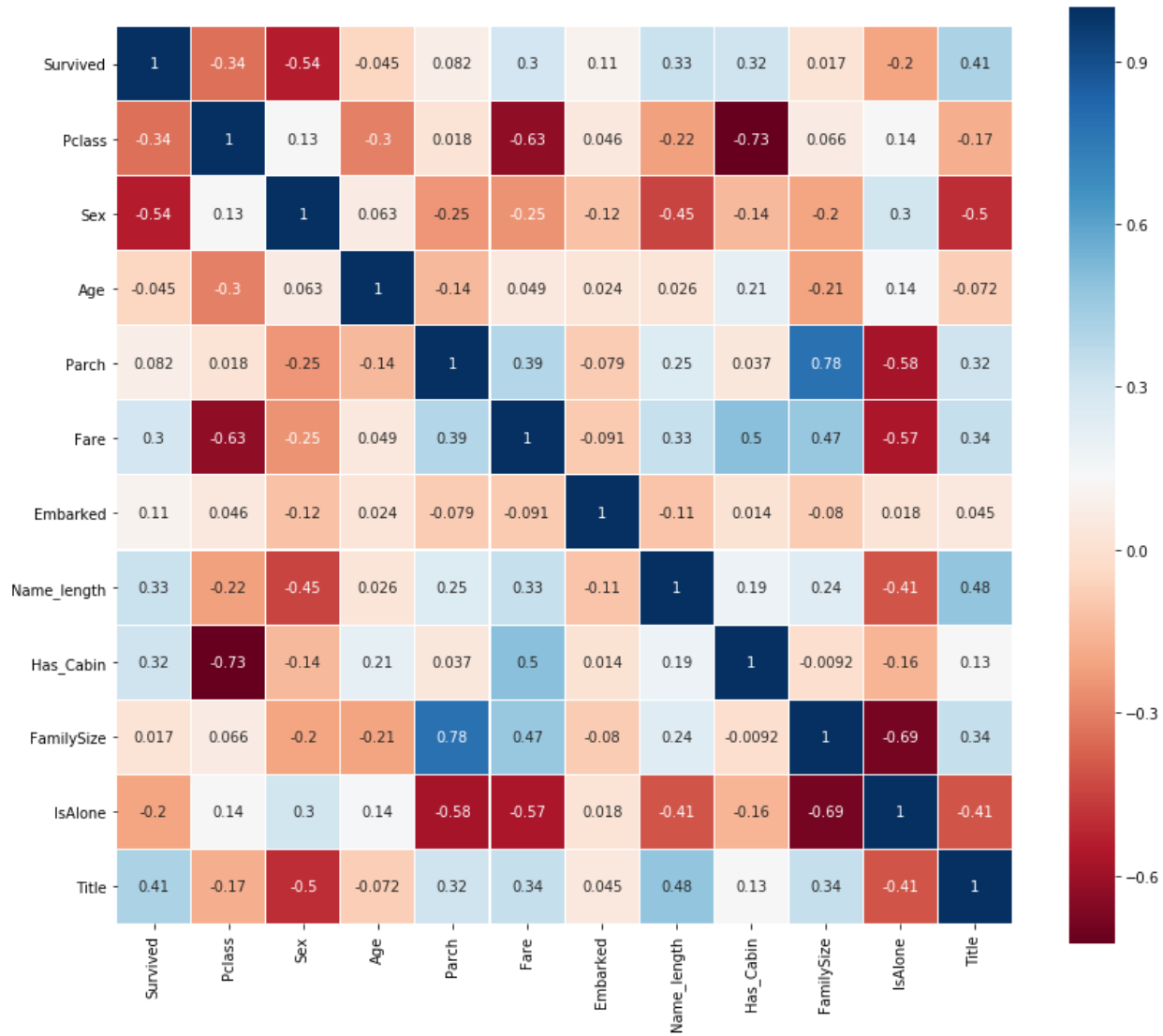
		Sex	female	male	All		
		Survived	0	1	0	1	
Embarked	C	Pclass					
		1	1	42	25	17	85
		2	0	7	8	2	17
	Q	3	8	15	33	10	66
		1	0	1	1	0	2
		2	0	2	1	0	3
	S	3	9	24	36	3	72
		1	2	46	51	28	127
		2	6	61	82	15	164
	All	3	55	33	231	34	353
			81	231	468	109	889



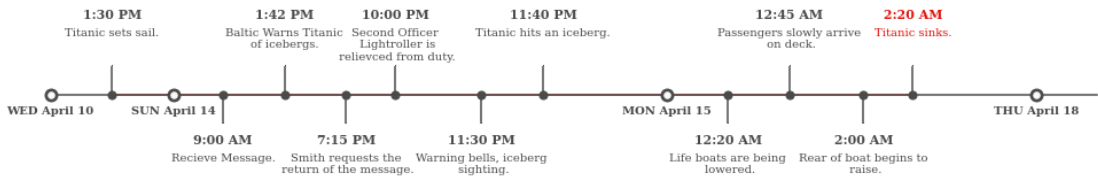


Pclass	1	2	3
SibSp			
0	137	120	351
1	71	55	83
2	5	8	15
3	3	1	12
4	0	0	18
5	0	0	5
8	0	0	7

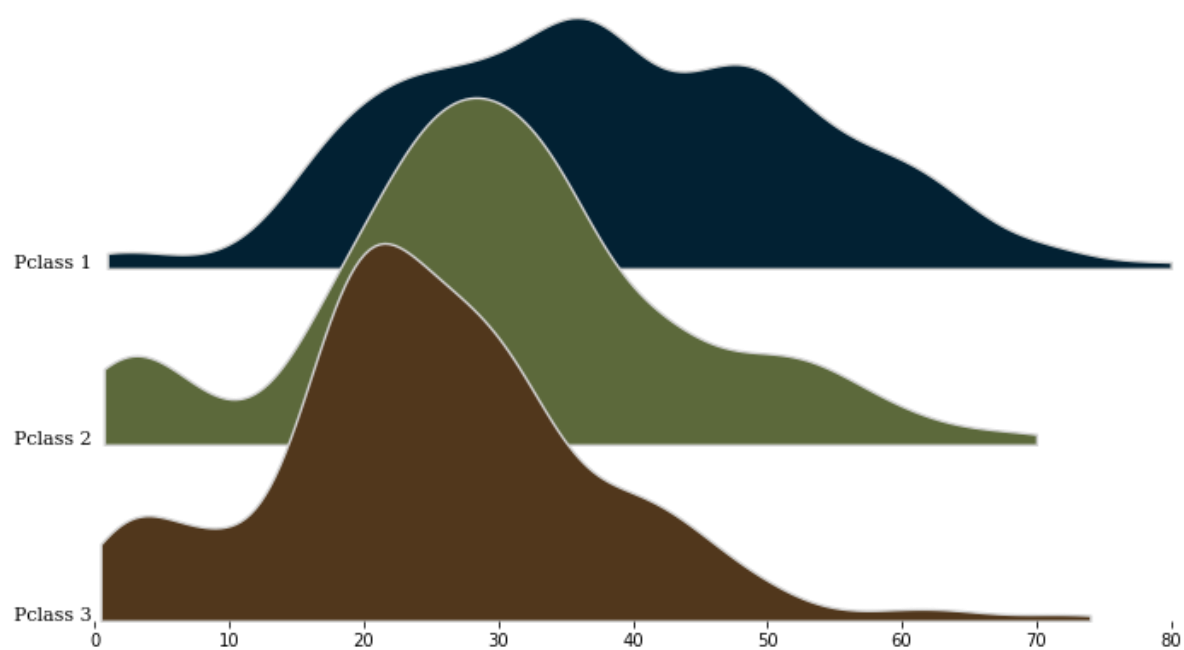
Pearson Correlation of Features



Titanic Timeline

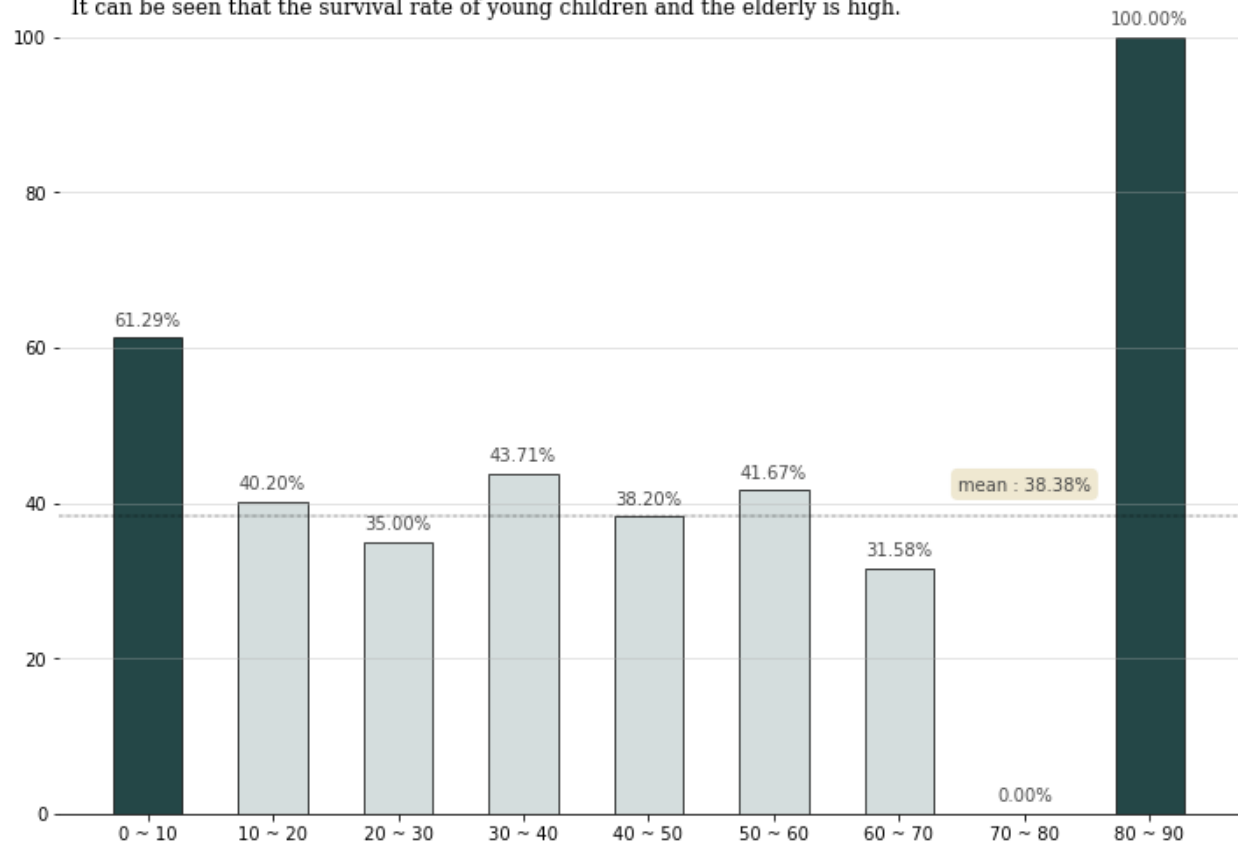


Age distribution by Pclass in Titanic

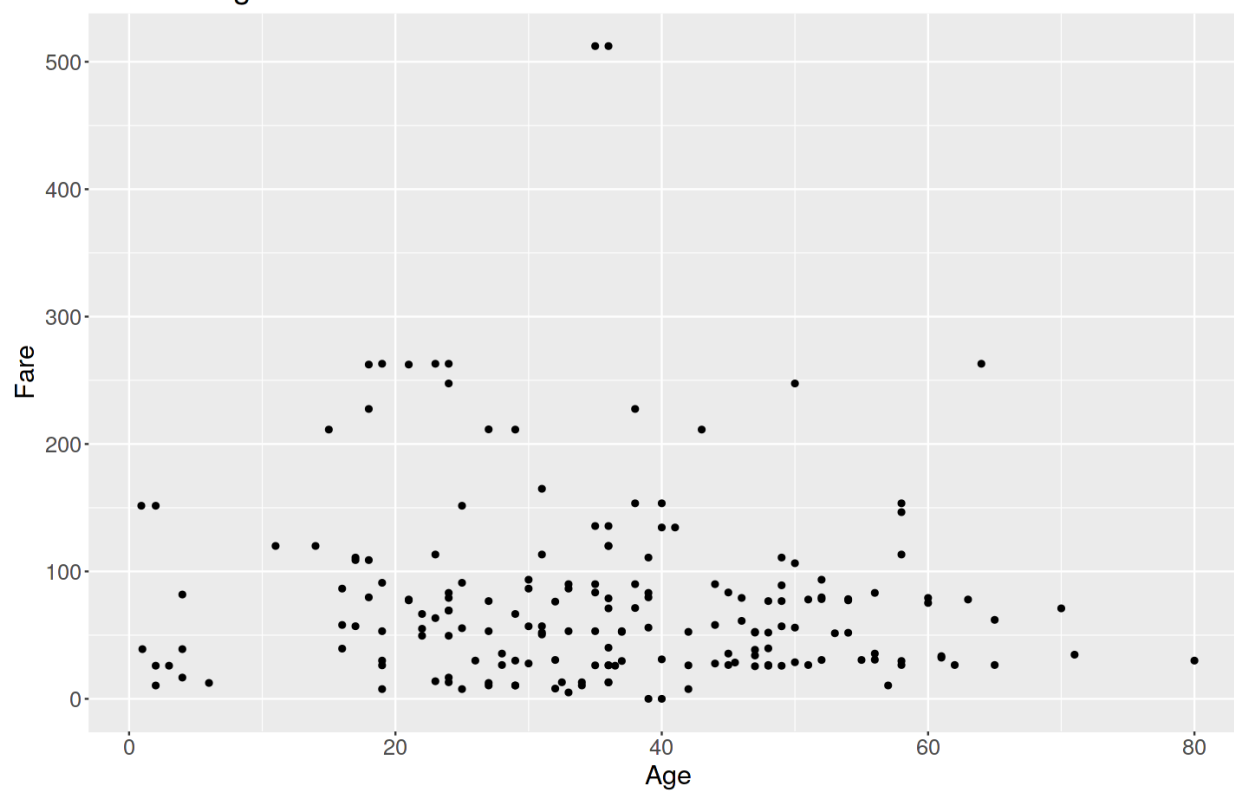


Age Band & Survival Rate

It can be seen that the survival rate of young children and the elderly is high.



Titanic - Age vs Fare



Dispersion of Age by Sex

